

WPR - 6

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 6) For Odd Semester of session 2026-2027

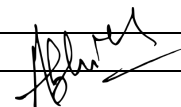
To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 569 2. 166 3. 191
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Applied to streaming token entropy and async sidecar proxy. Integrated Shannon entropy math with FastAPI SSE streaming.
Technical & Economical Feasibility	Feasible. Python 3.11+, PyTorch CUDA, FastAPI, Uvicorn, Streaming SSE, DeBERTa-v3 SLM (86M). Zero monetary cost (₹0).
Bill of Material	NA
Project Progress Schedule (PERT Chart)	• W1-W5: Setup, dataset curation, DeBERTa fine-tuning & evaluation (Completed) • W6-W8: Streaming entropy engine, early termination & FastAPI sidecar proxy (In Progress / Week 6) • W9-W12: NLI cross-encoder integration, comparative benchmarking & final IEEE paper
Design of critical components	1. Streaming Token Entropy Engine (<code>streaming_entropy_engine.py</code>) 2. Consecutive Spike Early Termination Controller 3. FastAPI Async Sidecar Proxy Server Prototype (<code>proxy_server.py</code>)
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	65% (Real-time streaming entropy calculator implemented, early termination triggers verified on hallucinated token sequences, FastAPI proxy server prototype built)
Result and Analysis	Verified sliding-window Shannon entropy $H(x)$ on streaming logits; confirmed early stream termination on sustained spikes (>1.20 bits), preventing delivery of hallucinated tokens.
Report writing	WPR Week 6 drafted.

Signature of students	1. Shourya Solanki Dhruv Raj Singh	2. Rachit Ryan Chug	3.
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Work done in this week (Week 6):

1. Streaming Token Entropy Module Implementation: Built `src/streaming\_entropy\_engine.py` to calculate Shannon Entropy $H(x) = -\sum(p * \log_2(p))$ over discrete token probability distributions in real time.
2. Consecutive Spike Early Termination Logic: Designed and verified threshold-based early termination triggering when uncertainty spikes past 1.20 bits across consecutive tokens, halting hallucinated text mid-sentence.
3. FastAPI Sidecar Proxy Server Prototype: Developed `src/proxy\_server.py` with `/v1/chat/completions` endpoint featuring asynchronous Pre-Scan security interception and Server-Sent Events (SSE) streaming token output.
4. Pre-Scan + Post-Scan Integration: Linked the fine-tuned DeBERTa-v3 classifier as pre-routing security middleware, returning 403 Forbidden with risk scores on detected prompt injection attempts.

To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory	
Performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 02/09/2026	 Signature of Guide